

Data Flow Diagram & User Stories

Date	15 July 2026
Team ID	XXXX
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	4 Marks

Data Flow Diagrams

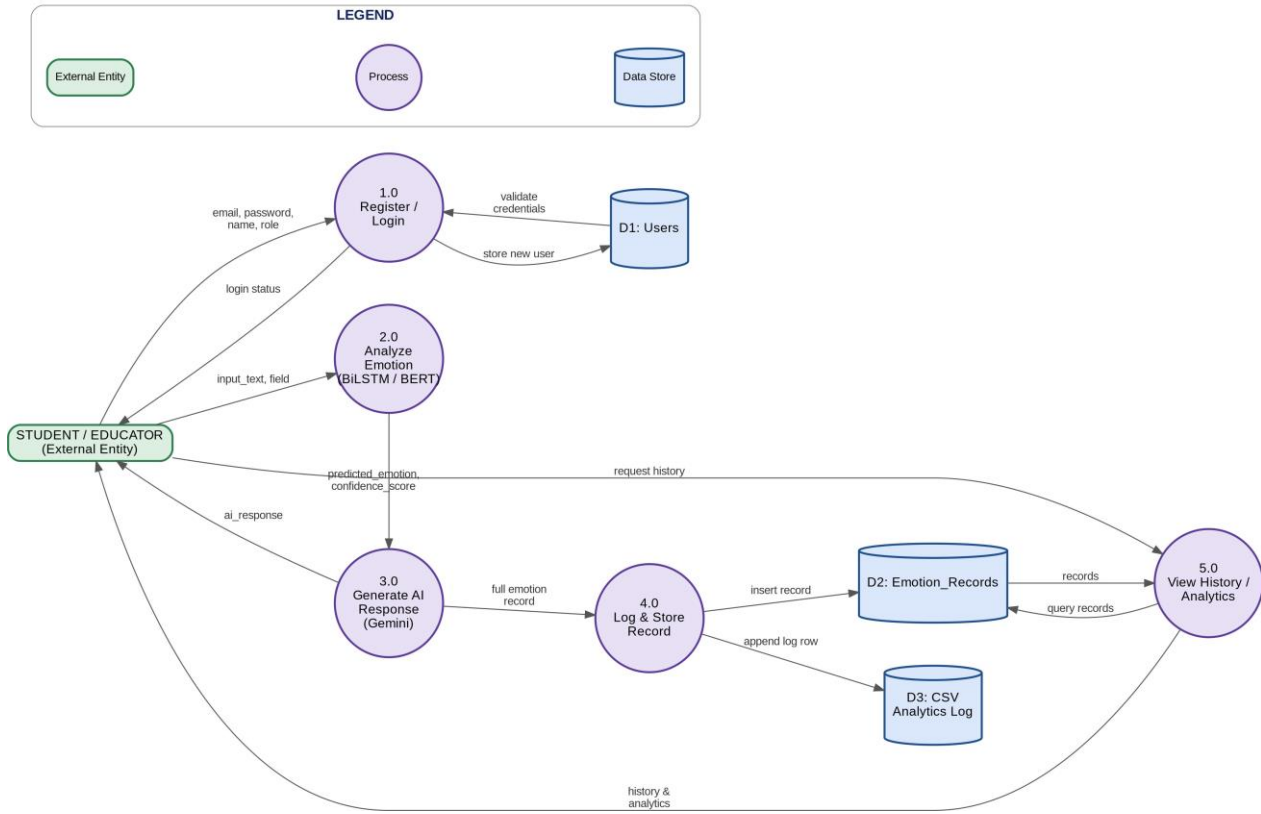
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Level 1 DFD - Emotion Detection & Learning Support Engine

The diagram below shows the five core processes of the system: user registration/login, emotion analysis of student input via the BiLSTM and BERT models, AI response generation via Gemini, logging of emotion records, and viewing of history/analytics - along with the data stores (Users, Emotion Records, CSV Analytics Log) each process reads from or writes to.

EMOTION DETECTION AND LEARNING SUPPORT ENGINE

Level 1 DFD



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Student	Registration & Access	USN-1	As a student, I can open the learning assistant application without needing to create a separate account.	I can access the field-selection and problem-input screen immediately	High	Sprint-1
Student	Emotion-Aware Input	USN-2	As a student, I can select my field of study and describe my learning problem in free text.	The field dropdown and text area are both available and required before submission	High	Sprint-1
Student	Emotion-Aware Input	USN-3	As a student, I can use one of the provided quick-example prompts instead of typing my own.	Clicking a quick example populates the text area with that example	Medium	Sprint-1
Student	Emotion Detection	USN-4	As a student, I want the system to detect my emotional state (Bored, Confident, Confused, Curious, Frustrated) from my text so I get relevant support.	Submitting a problem returns a predicted emotion with a confidence score	High	Sprint-1
Student	Emotion Detection	USN-5	As a student, I want to see when I'm feeling more than one emotion at once (e.g. Bored + Frustrated), not just a single forced label.	If a secondary emotion crosses the confidence threshold, it is displayed alongside the primary emotion	Medium	Sprint-2
Student	AI-Generated Guidance	USN-6	As a student, I want a supportive, field-specific response generated based on my detected emotion.	An AI-generated (or template fallback) response is shown, tailored to my field and emotion	High	Sprint-1
Student	Analytics & History	USN-7	As a student, I want to see my emotional journey over a session, so I can reflect on my own progress.	An analytics dashboard displays emotion distribution and an emotional-journey timeline	Medium	Sprint-2
Student	Analytics & History	USN-8	As a student, I can clear my session history if I want to start fresh.	Clicking "Clear History" resets the interaction log for the current session	Low	Sprint-2
Educator / TA	Model Comparison	USN-9	As an educator/TA, I want to compare BiLSTM and BERT predictions side-by-side to judge reliability.	Enabling "Show analysis details" displays both models' emotion breakdowns with confidence scores	Medium	Sprint-2
Administrator	Continuous Learning	USN-10	As an administrator, I want every interaction logged to CSV so the data can be used to improve future models.	Each submission appends a row to emotion_response_examples.csv and emotion_response_mapping.csv	Medium	Sprint-2